
Semantic Warp: Provenance-Conditioned Completion in Collective Memory

Anonymous Author(s)

Affiliation

Address

email

Abstract

Problem. In collective memory for multi-agent navigation, an agent can commit to — and reach — a target it has never observed: all supporting evidence arrived from peers. Folk culture calls this navigating by someone else’s song; we call it a *semantic warp*. Success-only and even stage-level metrics cannot see it, because no existing evaluation conditions completion on the *provenance* of the evidence behind the commitment. **Approach.** We define the warp event $W_\theta^* := M^* \wedge (\varphi \geq \theta)$, where foreignness φ is the peer share of the evidence mass *in the same weights the merge used to select the candidate*, frozen at lock time. Warp is not a fifth Q/R/M/C stage but an annotation that splits $P(C^*|M^*)$ into provenance strata. Because symbolic merges decompose by source, we obtain (i) a counterfactual attribution of warp value via deterministic replay with foreign evidence masked, (ii) a closed-form *warp distance law* from trust $\times$ staleness rules, and (iii) a decentralised *Warp Drive* protocol (optimistic lock + reservation + anti- M^* rollback) for the warp-collision pathology. **Findings.** On a 2,400-episode scarcity sweep, raw warp locks almost never complete directly ($P(C^*|W^*) = 0.004\text{--}0.020$ vs. $0.22\text{--}0.41$ for own locks; CIs disjoint), yet foreign evidence still carries value as transport: up to 51% of own-lock completions are warp-assisted chains. The distance law predicts intention-formation horizons tick-exactly, including a 6/6-exact hold-out with predictions registered before execution. Warp Drive turns the trade-off around: success rate rises at every cadence (disjoint CIs), and fast sharing with the protocol beats the best fixed-cadence baseline ($0.614\text{--}0.620$ vs. 0.583). A cross-session, cross-agent strict warp is replicated on an LLM substrate (10/10 layouts vs. 0/10 without foreign memory). **Claim.** We claim that collective memory must be evaluated not only by what information an agent retrieves, but by *whose evidence caused its commitment* and *under what temporal and coordination contract* that evidence remains actionable. **Thesis.** The value of collective memory lies not in shared information alone, but in the *provenance, validity horizon, and coordination contract* attached to that information. **Scope.** Grid-world substrates and one small text-world LLM demo; the contribution is a measurement layer, a falsifiable law, and a minimal protocol, not a general theory.

1 Introduction

In songline cultures, a traveller who knows the song can cross terrain they have never seen: navigational competence transfers without physical visitation. Multi-agent systems with shared memory perform the same act constantly — an agent commits to a water source, a goal region, or a tool location known only from a peer’s broadcast — yet no standard evaluation distinguishes this act from ordinary memory-driven navigation. Success metrics aggregate over it; stage-level diagnostics such

37 as Q/R/M/C count the commitment (M^*) and the completion (C^*) but not *whose evidence* backed
38 them.

39 This paper makes that act a first-class measured event. A *semantic warp* W^* is a target materialisation
40 whose supporting evidence mass is (almost) entirely foreign, and *warp completion* $P(C^*|W^*)$ asks
41 whether someone else’s song actually gets you there. Three properties of symbolic collective memory
42 make the construction sharp, and none of them holds for opaque vector memories:

- 43 1. **Merges decompose by source.** Foreignness $\varphi \in [0, 1]$ is computed from the same per-source
44 weights (trust, staleness, support) the merge itself used to rank the candidate — it is a reading of
45 the decision, not a new heuristic.
- 46 2. **Replays are deterministic.** Re-running an episode with foreign contributions masked at merge
47 yields a *counterfactual* warp gain per episode, not a correlation.
- 48 3. **Sharing rules have closed forms.** Trust and staleness decay imply a *warp distance law*: an
49 explicit horizon beyond which foreign evidence can no longer form intention — falsifiable, and
50 falsifiably different across architectures.

51 Our measurements then force a reframing. Raw warp is nearly worthless at the point of completion:
52 under scarcity, warp locks complete an order of magnitude less often than own-evidence locks
53 (§4), because concurrent recipients of the same broadcast race for the same target and most must
54 lose. Yet the same foreign evidence is demonstrably valuable as *transport*: up to half of all own-
55 evidence completions are chains that began with a failed warp lock which carried the agent into
56 self-confirmation range (§4.1). And a lightweight decentralised contract — a reservation piggybacked
57 on the existing broadcast plus an anti- M^* rollback in the sense of Time Warp simulation Jefferson
58 (1985) — recovers what raw warp loses: with the protocol, *fast* sharing becomes the best-performing
59 regime on unconditional success rate (§6). Hence the thesis in one sentence: **the value of collective**
60 **memory is not the information; it is the coordination contract attached to the information.**

61 **What we claim.** We claim that provenance is a necessary evaluation *and* control dimension for
62 collective memory. An agent may commit to a target it has never observed using evidence obtained
63 entirely from peers, yet this dependency is invisible both to success metrics and to stage-level
64 evaluation without source attribution; we make it observable and counterfactually testable. The paper
65 supports four empirical claims. **(EC1)** Raw foreign locks have substantially lower direct completion
66 than own-evidence locks, but foreign evidence produces positive downstream value through mixed-
67 provenance lock chains. **(EC2)** Under the tested exponential trust–staleness gate, the latest tick at
68 which foreign evidence can form an intention is predicted exactly by the closed-form warp distance
69 law, including the pre-registered 6/6 hold-out. **(EC3)** The completion loss caused by fast sharing is
70 concentrated in foreign-evidence commitments and is attributable to target contention rather than to
71 inaccurate evidence. **(EC4)** A decentralised reservation-and-rollback contract increases success at
72 every tested fast cadence and outperforms the best fixed-cadence baseline.

73 Contributions.

- 74 1. The first provenance-conditioned completion metric for collective memory: W_{θ}^* , φ
75 frozen at lock time, the decomposition $P(C^*|M^*) = P(C^*|M^*, \varphi < \theta) P(\varphi < \theta|M^*) +$
76 $P(C^*|W^*) P(W^*|M^*)$, and the counterfactual warp gain via source-masked deterministic
77 replay (§3–§4).
- 78 2. The *warp distance law* $\text{age}_{\max}(\tau_i) = \alpha^{-1} \ln(\tau_i c / \tau)$ derived from the memory’s own
79 trust×staleness rules, validated tick-exactly including a registered-prediction hold-out (6/6
80 exact, one cell where a pruning horizon must undercut the exponential gate — and does) and an
81 architecture-level shape difference: fixed-cadence peer memory has *no* gate (§5).
- 82 3. The *Warp Drive* protocol: optimistic warp lock, reservations carried on the existing broadcast
83 cadence (no coordinator), and anti- M^* rollback with exponential backoff. It resolves the
84 known misled-agent pathology preventively, bounds rollback latency by $\approx K$ ticks, and lifts
85 unconditional success rate above the best fixed-cadence baseline (§6).
- 86 4. A cross-session, cross-agent strict warp on an LLM substrate: an agent that has never observed
87 a room solves a task there purely on another agent’s previous-session memory, replicated 10/10
88 layouts against a 0/10 control (§7).

89 **Relation to the companion papers.** The Q/R/M/C framework paper treats stage rates as the
 90 measurement instrument; the collective-memory paper establishes the architectures and the cadence
 91 trade-off. This paper adds the provenance axis on top of both: the same logs, one new annotation,
 92 and the mechanisms the annotation exposes.

93 2 Setting

94 **Substrate and task.** N agents explore a 12×10 grid with M semantic water targets ($N > M$:
 95 scarcity) and hazards. Each agent maintains symbolic place memory (tags, confidences, support,
 96 freshness). Four sharing architectures over the identical substrate and planner: independent (no
 97 communication), shared (single event bus), centralized (consensus aggregator), peer (broadcast
 98 every K ticks, asymmetric trust), plus CSM (peer + explicit merge/trust/staleness rules with decay
 99 rate $\alpha=0.05$ and inclusion threshold $\tau=0.30$).

100 **Q/R/M/C instrumentation.** Per tick and per agent we log query formation (Q), retrieval satisfaction
 101 (R), target materialisation as the planner’s lock (M^*), and completion (C^*). Episode-level conditional
 102 rates follow the companion definition. All warp measurements below are annotations of these M^*
 103 events; no planner or memory invariant is modified.

104 3 The warp event

105 **Provenance and foreignness.** Every evidence unit entering a merge carries its source. For the
 106 candidate m the planner locks,

$$\varphi(m) = \frac{\sum_{e \in \text{supp}(m), \pi(e) \neq \text{self}} w(e)}{\sum_{e \in \text{supp}(m)} w(e)},$$

107 where $w(e)$ are *the merge’s own weights*: trust $\cdot \log(1+\text{support})$ for peer/centralized consensus,
 108 trust $\cdot e^{-\alpha \text{ age}}$ $\cdot$ conf for CSM, observation masses for the shared bus. For peer and centralized
 109 architectures φ is reconstructed exactly from the per-contribution fields the merge already exports;
 110 the measurement layer reads state and never writes it.

111 **Freezing rule.** φ is fixed at the moment of the lock. If the agent later confirms the place with its
 112 own eyes, φ is *not* recomputed — otherwise every successful warp would dilute itself on approach
 113 and $P(C^*|W^*)$ would be unmeasurable by construction.

114 **Warp events and strata.** $W_\theta^* := M^* \wedge (\varphi \geq \theta)$ with $\theta=1$ (strict: never self-observed, cross-
 115 checked against an observation ledger) and $\theta=0.8$ (soft). Warp share $P(W^*|M^*)$ is a structural
 116 descriptor (identically 0 for independent); warp completion $P(C^*|W^*)$ is the new quantity. The
 117 chain-rule decomposition in §1 makes warp an exact refinement of $P(C^*|M^*)$, not a new stage:
 118 identifiability of Q/R/M/C is untouched.

119 **Regression anchor.** The known misled-agent episode of the architecture comparison (agent A locks
 120 (3, 8) known only from C; C claims it first; A strands) is annotated by the instrumentation as $\varphi = 1.0$
 121 strict W^* with collision — the pathology the framework must see, and the protocol of §6 must fix.

122 4 Raw warp rarely completes — but transports

123 **Design.** 2,400 episodes: $(N, M) \in \{(3, 2), (5, 3), (8, 5), (4, 4), (6, 2)\} \times \text{three layouts} \times$
 124 $\{\text{peer-}K2, \text{peer-}K8, \text{CSM-}K8, \text{shared}\} \times 20 \text{ seeds} \times \{\text{full, foreign-masked}\}$ with identical seeds.
 125 Unit of analysis: lock event; CIs by episode-cluster bootstrap.

126 Figure 1 shows the cadence view on a dedicated peer sweep ($K \in \{1, 2, 4, 8, 16\}$): warp share is
 127 what fast sharing buys, and it is precisely the stratum that does not complete. Table 1 gives the
 128 architecture view.

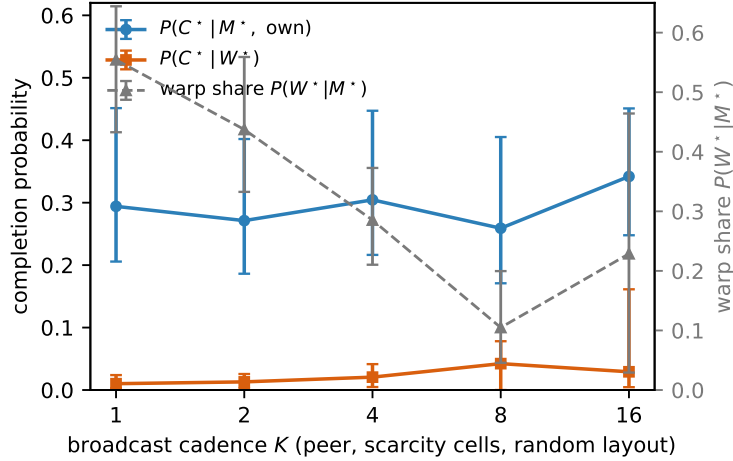


Figure 1: φ -stratified completion vs. broadcast cadence K (dedicated peer sweep: scarcity cells, random layouts, 20 seeds, episode-cluster bootstrap CIs). As K shrinks, the warp share $P(W^*|M^*)$ grows from 0.11 to 0.55 while $P(C^*|W^*)$ stays below 0.05 and the own-stratum rate stays flat: the completion collapse of fast sharing is concentrated in the W^* stratum, giving the companion paper’s bottleneck shift its mechanistic reading. The share uptick at $K=16$ (broadcasts are rare, but agents also discover little on their own) is reported as observed.

Table 1: Provenance strata of completion under scarcity ($N > M$), full runs. Warp locks complete an order of magnitude less often than own-evidence locks; all CI pairs are disjoint. (H-W1 confirmed.)

Architecture	$P(C^* W_{0.8}^*)$	n_{W^*}	$P(C^* M^*, \text{own})$	n_{own}
peer- $K=2$	0.004 [0.001, 0.008]	1,437	0.223 [0.186, 0.275]	2,738
peer- $K=8$	0.020 [0.000, 0.038]	149	0.389 [0.329, 0.446]	1,734
CSM- $K=8$	0.017 [0.006, 0.044]	637	0.414 [0.368, 0.453]	1,684
shared	0.004 [0.002, 0.008]	2,099	0.279 [0.237, 0.332]	2,250

4.1 Warp as transport: lock chains

Direct completion understates the value of foreign evidence. A *warp-assisted own completion* is a completed own-evidence lock preceded (same agent, same episode) by a soft- W^* lock that was dropped: the warp failed as a destination but succeeded as a ride.

Table 2: Lock-chain analysis (random layouts, full runs). The first W^* lock in a chain is made from ≈ 8 –10 cells away; the completing own lock from 2.0 cells — exactly the observation radius, i.e. the self-confirmation point.

Arm	own C^*	warp-assisted (share)	$r(\text{first } W^*) \rightarrow r(\text{own})$
shared	248	127 (51.2%)	9.5 $\rightarrow$ 2.0
peer- $K=2$	223	99 (44.4%)	9.7 $\rightarrow$ 2.0
CSM- $K=8$	279	13 (4.7%)	5.8 $\rightarrow$ 2.0
peer- $K=8$	246	7 (2.8%)	8.4 $\rightarrow$ 2.0

This resolves an apparent paradox in the counterfactual data: the shared bus at moderate scarcity gains +10.8 ticks [+2.2, +20.9] from foreign evidence although its direct $P(C^*|W^*)$ is 0.004. The gain flows through chains; the freezing rule books the completion honestly in the own stratum, and the chain analysis recovers the warp’s contribution as a separate measured quantity.

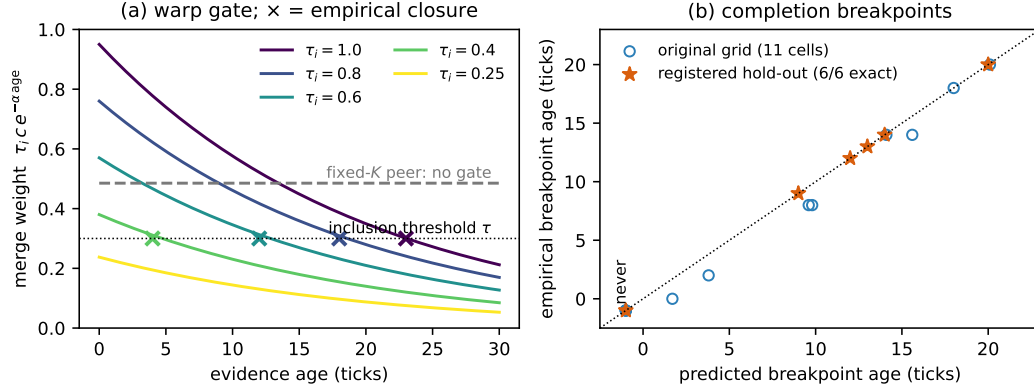


Figure 2: The warp distance law. **(a)** Foreign-evidence merge weight under trust $\times$ staleness decay; $\times$ marks the empirically measured last age at which the traveller’s query still exposes the target — each sits on its curve’s crossing of the inclusion threshold τ . Fixed-cadence peer memory has no staleness term (dashed): its gate never closes; the architectures differ in the *shape* of the curve. **(b)** Predicted vs. empirical completion breakpoints in full navigation episodes: original grid (circles; age sampled in steps of 2, hence points at most one grid step below the diagonal) and the registered hold-out (stars; age step 1, predictions written to disk before execution, 6/6 exact, including the never-cell at $(-1, -1)$ and the cell where the pruning horizon undercuts the exponential gate).

137 4.2 Counterfactual warp gain

138 Masking foreign evidence at merge under the same seed gives $WG = t_{\text{cens}}(\text{masked}) - t_{\text{cens}}(\text{full})$ per
 139 episode. The sign map over (layout, scarcity $\rho = N/M$) is regime-dependent: positive for the shared
 140 bus at unpredictable layouts and moderate scarcity ($+10.8$ at $\rho=1.5$), negative for fast peer sharing
 141 (peer- K^2 random: $-7.1 [-13.5, -1.3]$), ≈ 0 at hard scarcity. A per-agent mask (one agent masked,
 142 others share normally) confirms the sign at the individual level: $WG_i = -10.25 [-20.91, -0.16]$
 143 ticks on the fast-share scarcity cell, with spillover on unmasked agents ≈ 0 ($-0.62 [-3.06, +1.91]$).
 144 Raw warp *hurts* its recipient precisely where sharing is fastest — the collision cost exceeds the
 145 information value.

146 **Exploratory: collision pressure.** Co-lock pressure (other agents locked on the same target)
 147 correlates with warp failure with the predicted sign but negligible magnitude (Spearman $+0.033$,
 148 $p=0.018$, $n=5,117$); we report it as exploratory rather than as a confirmed predictor.

149 5 The warp distance law

150 CSM includes foreign evidence with weight $\tau_i e^{-\alpha \text{age}} c$ (trust τ_i , confidence $c=0.95$) and inclusion
 151 threshold $\tau=0.30$. Foreign evidence therefore stops forming intention at

$$\text{age}_{\max}(\tau_i) = \frac{1}{\alpha} \ln(\tau_i c / \tau), \quad \text{defined for } \tau_i c > \tau.$$

152 **Witness–traveller protocol.** A witness observes water once and goes silent; a traveller who has
 153 never seen it receives the snapshot at a parameterised age. Every lock the traveller makes is a strict
 154 W^* by construction. Figure 2 summarises both experiments.

155 **Completion breakpoint.** In full navigation episodes the feasibility inequality acquires one discrete
 156 refinement: the traveller self-confirms the target once it is within the observation radius, so the foreign
 157 gate only has to survive until tick $t_{\text{gate}} = d - r_{\text{obs}} - 1$, giving breakpoint $\text{bp} = \text{age}_{\max} - t_{\text{gate}}$;
 158 snapshot pruning executes on broadcast ticks and adds the bound $10K - t_{\text{bcast}}$. All constants of
 159 the law are unchanged. All 11 CSM cells of the original grid match within sampling resolution;
 160 completion curves are monotone steps; the trust-flip zeroes W^* events discretely.

161 The law is thus predictive out of sample, tick-exactly: the CSM rules really are the mechanism
 162 that forms (and un-forms) intention. It also reinterprets the companion result that CSM dominates

Table 3: Gate curve: largest evidence age at which the traveller’s query still exposes the target. Fixed-cadence peer memory has no staleness term: the gate never closes (checked to age 120 at all trusts) — the architectures differ in the *shape* of the curve, not just its level.

trust τ_i	empirical age _{max}	predicted
1.00	23	23.05
0.80	18	18.59
0.60	12	12.84
0.40	4	4.73
0.25	gate closed	0 (trust-flip)

Table 4: Registered-prediction hold-out: six cells absent from the original grid, predictions written to disk *before* any episode, age grid step 1. All six are exact integer matches, including the never-cell and the cell where the pruning horizon must undercut the exponential gate.

Cell	binding bound	predicted	empirical
$K=8, \tau_i=0.9, d=9$	gate	14	14
$K=8, \tau_i=0.7, d=9$	gate	9	9
$K=8, \tau_i=0.5, d=15$	gate	never	never
$K=4, \tau_i=1.0, d=6$	gate	20	20
$K=2, \tau_i=1.0, d=12$	<i>pruning</i>	12	12
$K=2, \tau_i=0.8, d=8$	gate	13	13

fixed-cadence peers: trust $\times$ staleness is a *warp gate* — CSM does not share better, it *refuses to warp on a stale or untrusted song*.

Portability to continuous dynamics. All three registered predictions survive on a continuous VMAS substrate through the same continuous $\rightarrow$ grid bridge used by the companion’s portability test: (i) the provenance strata stay an order of magnitude apart ($P(C^*|W^*) = 0.012$ [0.000, 0.042] vs. 0.562 [0.457, 0.709] at $K=1$); (ii) warp share falls monotonically with cadence ($0.723 \rightarrow 0.366 \rightarrow 0.000$ over $K=1/4/16$); (iii) the law becomes *metric*: with the travel-to-sight time calibrated once on a fresh-evidence episode, the predicted breakpoints match within the sampling grid, and all cells beyond the metric warp radius ($v \cdot \text{age}_{\text{max}}$ shorter than the distance) are predicted *never* and observed *never*. One genuinely continuous finding: inertia is a physical warp-leak channel — an agent can coast into sighting range after the gate closes — removed by a no-lock-then-brake controller that restores the grid semantics.

6 The Warp Drive protocol

Warp collision — locking on foreign evidence for a target someone else will claim — is the dominant failure of fast sharing. Warp Drive adds three primitives, all decentralised:

1. **Optimistic warp lock** — locks on foreign evidence remain allowed immediately (planner unchanged).
2. **Warp reservation** — at the owner’s next broadcast tick the reservation rides on the ordinary snapshot (same cadence K , no coordinator, no extra channel); receivers suppress reserved targets in their queries. Priority: (lock tick, distance at lock, agent id).
3. **Anti- M^* rollback** (Time Warp semantics Jefferson (1985)) — an agent holding a lock dominated by a foreign reservation retracts it: the planner re-queries, the target enters exponential backoff ($4 \rightarrow 8 \rightarrow \dots \rightarrow 32$ ticks) with an agent-id tie-break against oscillation. The event log stays append-only.

Decentralisation check. Driving the protocol by direct per-agent calls — no runtime, no environment — produces the same retraction decisions from each agent’s own inbox alone.

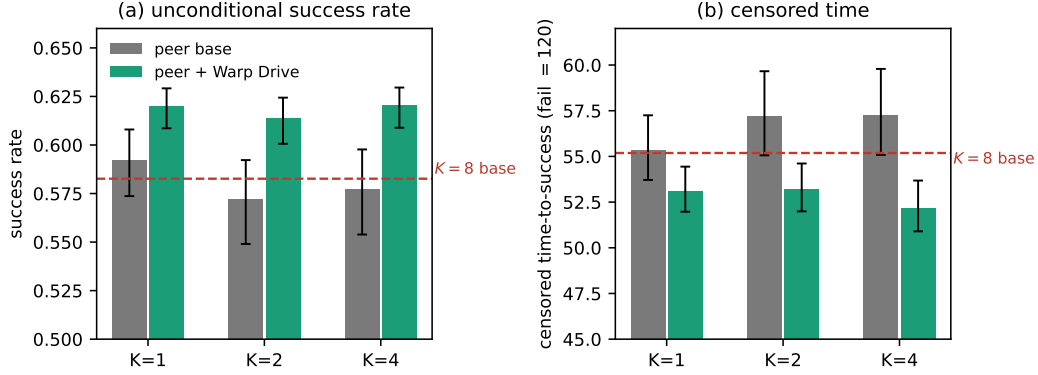


Figure 3: Warp Drive on unconditional metrics (840 episodes; bootstrap CIs). **(a)** Success rate rises at every cadence with disjoint CIs, and every fast- K protocol arm exceeds the $K=8$ sweet-spot baseline (dashed). **(b)** Censored time-to-success falls correspondingly. The claim that the protocol removes the main argument against fast sharing rests on these unconditional quantities, not on the (denominator-sensitive) conditional rates.

189 **Misled-A anchor.** With Warp Drive the pathological lock is resolved *preventively*: C’s reservation
 190 arrives before A ever locks (3, 8); A completes on another target. In the sweep, rollback latency of
 191 dominated locks is finite and $\approx K$: 1.0/1.5/2.7 ticks at $K=1/2/4$ (baseline: until end of episode).

Table 5: Unconditional results, 840 episodes (scarcity cells $\times$ random/asymmetric layouts $\times$ 20 seeds). Success rate rises under Warp Drive at every cadence with disjoint CIs, censored time falls, and fast sharing with the protocol beats the $K=8$ sweet-spot baseline (Figure 3). Lock census shows the conditional-metric denominator shrinking — which is why the claim is stated on unconditional quantities.

Arm	success rate	t_{cens} (fail = 120)	lock census
$K=1$ base	0.592 [0.574, 0.608]	55.3 [53.7, 57.2]	0.986
$K=1$ +WD	0.620 [0.609, 0.629]	53.1 [52.0, 54.4]	0.985
$K=2$ base	0.572 [0.549, 0.592]	57.2 [55.1, 59.7]	0.973
$K=2$ +WD	0.614 [0.601, 0.624]	53.2 [52.0, 54.6]	0.926
$K=4$ base	0.577 [0.554, 0.598]	57.3 [55.1, 59.8]	0.944
$K=4$ +WD	0.620 [0.609, 0.630]	52.2 [50.9, 53.7]	0.739
$K=8$ base	0.583 [0.558, 0.604]	55.2 [52.9, 57.9]	0.682

192 On the conditional axis, recovery is measured against the scarcity ceiling $\min(m, M)/m$ (no protocol
 193 conjures extra water): 0.90/2.15/6.6 of the recoverable collapse at $K=1/2/4$; values above 1 arise
 194 because reservation suppression removes doomed locks from the M^* population. Together with
 195 Table 2 this closes the arc: raw warp fails at completion (§4); with a coordination contract attached,
 196 warp at fast cadence becomes the best regime. The cadence sweet spot of the companion paper is
 197 reinterpreted: moderate K was never about information freshness — it was an implicit rate limiter on
 198 warp collisions, and an explicit contract does the job better.

199 7 Cross-session warp on an LLM substrate

200 Finally, the strict warp crosses both an agent boundary and a session boundary on a language
 201 substrate. In session 1, agent A (a local LLM tag extractor over natural-language observations;
 202 Ollama llama3.1, deterministic caching, zero API cost) sweeps a three-room text household and its
 203 symbolic memory is dumped to disk. In session 2, agent B — a fresh agent in a fresh environment
 204 instance, with the stock directional oracle removed so that long-range knowledge can only come from
 205 memory — loads A’s dump as foreign provenance and solves “bring the apple to the table.” Every
 206 lock B makes on the apple or the table is $\varphi=1.0$ strict W^* ; execution is layered (the LLM makes the

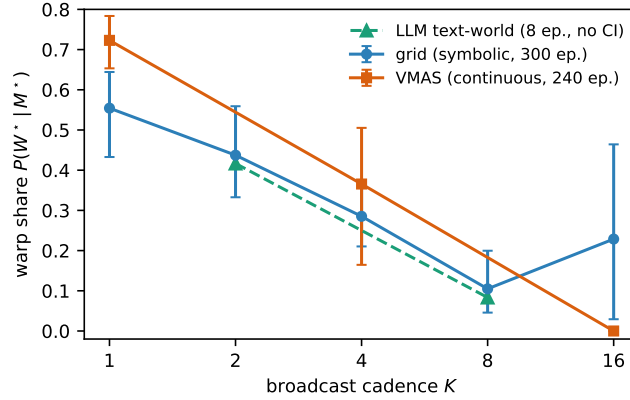


Figure 4: One quantity, three substrates: warp share $P(W^*|M^*)$ vs. broadcast cadence K on the symbolic grid, the continuous VMAS scenario, and the LLM text-world. The curves nearly coincide — how much of a collective’s materialisation runs on foreign evidence is set by the sharing cadence, not by the substrate. (LLM arm: 8 episodes, point estimates only.)

207 semantic decisions — which remembered place to pursue, when to take/put — while locomotion to
 208 the locked waypoint is a deterministic motor primitive, matching the substrate’s planner stack).

Table 6: Cross-session warp over 10 randomised layouts. Completion is attributed at the environment’s affordance radius (take/put act within Manhattan distance 1, and the witness tags anchors on within-reach cells).

	success	mean t_{succ}	layouts with completed strict W^*
warp (A’s memory)	10/10	8.3	10/10
control (empty memory)	0/10	—	—

209 To our knowledge this is the first documented cross-session, cross-agent strict semantic warp on an
 210 LLM substrate. The 10/10 vs. 0/10 contrast shows the warp is load-bearing, not decorative.

211 **Scaling to a live collective.** With $N=3$ LLM agents sharing by peer broadcast under scarcity ($M=2$
 212 apples, four layouts, $K \in \{2, 8\}$), the full warp phenomenology reproduces on the language substrate
 213 (all predictions registered before the runs): strict W^* locks occur; warp share falls with cadence (0.417
 214 at $K=2$ vs. 0.083 at $K=8$ — the same shape as the grid 0.554 $\rightarrow$ 0.105 and VMAS 0.723 $\rightarrow$ 0.000);
 215 warp collisions appear in 7/8 episodes. A paired comparison against the no-communication baseline
 216 on identical layouts shows raw sharing *lowering* success from 2/3 to 1/3 per episode — the collision
 217 cost exceeds the information value on the LLM substrate exactly as in §4.2. Attaching the unchanged
 218 Warp Drive protocol to the LLM collective closes the arc on language too: pooled success rises from
 219 0.333 to 0.417 at both cadences with rollback latencies again $\approx K$ (1 tick at $K=2$, 7 at $K=8$). The
 220 recovery is partial ($\approx 25\%$ of the collapse, vs. 90–100% on the grid): on a language substrate the
 221 collision pathology is compounded by semantic-level losses (hallucinated anchor tags, noisy locks)
 222 that no reservation can cure — a measurable decomposition we leave to future work. The distance
 223 law also survives LLM-produced evidence: with the gate constant c set to the confidence the extractor
 224 actually emitted ($c_{\text{meas}}=0.900$), all four registered breakpoints land within one tick of prediction.

225 Two interface findings transfer to any LLM-over-symbolic-memory design: the LLM query former
 226 free-generates retrieval tags outside the memory’s vocabulary (requiring an any-of gate), and small
 227 local models are reliable at semantic commitment but not at coordinate arithmetic or systematic
 228 exploration (requiring the layered execution above and an explicit exploration tier — without one,
 229 discovery never happens and the warp has nothing to feed on).

230 Figure 4 overlays the warp-share-cadence curve across all three substrates: the quantity behaves as a
 231 substrate-free property of the sharing architecture.

8 Related work

Shared memory in agent collectives. Generative Agents Park et al. (2023) and multi-agent LLM frameworks share memory streams or context but expose no lock interface and no per-source decomposition of a merged belief, so provenance-conditioned completion is not measurable there. Vector memories (e.g. MemGPT Packer et al. (2023)) merge by embedding arithmetic; the merge is not source-decomposable, which blocks both φ and the masked-replay counterfactual.

Emergent communication. CommNet Sukhbaatar et al. (2016) and TarMAC Das et al. (2019) learn continuous channels; the companion paper shows such channels collapse stage observability. Warp requires an explicit materialisation event, which these architectures do not expose.

Stigmergy. Pheromone-style coordination is an *anonymous* warp: evidence transfers without provenance. Our results quantify what anonymity costs (shared-bus warp share is maximal while its warp completion is minimal) and what provenance buys (trust gates, reservations, rollback).

Optimistic concurrency. The rollback primitive is Time Warp Jefferson (1985) transplanted from parallel simulation to memory-driven navigation: locks are optimistic computations, occupancy evidence is a straggler message, anti- M^* is the anti-message. To our knowledge this transplant is new.

9 Limitations and honest negatives

(0) All experiments assume a SHARED coordinate frame: place identity is coordinate proximity with semantics as a tie-breaker inside a spatial radius; the cross-agent place-correspondence problem is bracketed, not solved. (A frame-free extension — fingerprint-based semantic identity with landmark-consensus frame recovery, failing closed under perceptual aliasing where the coordinate assumption fails open — is implemented and validated separately and is out of scope here.) (1) Grid worlds, a continuous VMAS scenario, and one small text world; no photorealistic substrate. (2) The co-lock-pressure predictor survived only as exploratory (§4.2). (3) The warp-gain sign map is regime-dependent rather than the clean interior structure the pre-registered hypothesis expected; deterministic symmetric/asymmetric layouts carry little seed variance, so the statistical weight rests on random layouts. (4) The distance-law refinement (self-confirmation horizon, pruning schedule) was formulated after seeing the original grid — mitigated, but not erased, by the registered hold-out. (5) The LLM demo uses one model family and a deliberately small environment; it is a demonstration of measurability, not a benchmark. (6) All deviations from the pre-registered design are listed in Appendix A.

10 Conclusion

Warp turns “agents share memory” from an architecture property into a measured, causally attributable event with a falsifiable distance law and a protocol that acts on it. The measurements dissolve the information-centric reading of collective memory: raw foreign evidence mostly fails its recipient at completion and pays off as transport, while a two-line coordination contract — reserve on lock, roll back on domination — makes the fastest sharing regime also the best one. Provenance, not volume, is the control surface that matters.

Outlook: the contract as one function among several. The contract perspective suggests a reading we state as a falsifiable programme rather than a conclusion: memory is not one mechanism but several functions with distinct failure signatures. The trust $\times$ staleness gate studied here implements *memory-for-trust* (ablate it and completion collapses through stale and poisoned commits — the trust-flip and the raw-exchange negative are exactly that signature); it is distinct from *memory-for-return* (own consolidation; its ablation starves candidate generation) and from *memory-for-telling* (what is worth broadcasting at all; its ablation leaves receivers’ retrieval unimproved on tags they undersample). Under this reading the right validity criterion for shared memory is not direct verification of content but *ecological viability*: whether a commitment formed on foreign evidence can still be carried to completion under the contract — which is precisely the quantity $P(C^* \mid W^*)$ and its law make measurable.

A Deviations from the pre-registered design

1. Reservation tie-break extended to (lock tick, *distance at lock*, agent id): a pure id tie-break hands simultaneous locks to a distant agent over the target’s near-discoverer.
2. Protocol recovery is measured against the scarcity ceiling $\min(m, M)/m$ rather than the $K=8$ reference, separating structural from coordination collapse; unconditional metrics are reported alongside (Table 5).
3. Completion breakpoints use the discrete-time refinement of §5 (law constants unchanged; hold-out registered).
4. Collision pressure is operationalised as co-lock count rather than co-recipients (structurally constant $N-2$ under broadcast).
5. LLM demo: layered execution (LLM semantics, deterministic locomotion); completion attributed at the affordance radius.

B Reproducibility

All experiments run on CPU in minutes-to-tens-of-minutes; the LLM demo uses a local Ollama model with deterministic caching.

```
PYTHONPATH=. python experiments/warp/exp_warp_anchor.py
PYTHONPATH=. python experiments/warp/exp_warp_gain.py --seeds 20
PYTHONPATH=. python experiments/warp/analyze_warp_chains.py
PYTHONPATH=. python experiments/warp/exp_warp_gain_individual.py
PYTHONPATH=. python experiments/warp/exp_warp_age_law.py
PYTHONPATH=. python experiments/warp/exp_warp_age_law_holdout.py
PYTHONPATH=. python experiments/warp/exp_warp_drive.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_cross_session.py --layouts 10
```

C Full counterfactual sign map

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